Owen Kim

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Education

University of Waterloo Sep. 2024 – May 2029

BASc Biomedical Engineering – GPA: 3.91/4.0 (89% Cumulative Average)

Relevant Courses: Systems and Signals, Circuits, Data Structures and Algorithms, Statistics

Experience

The Hospital for Sick Children Jan. 2026 – Present

R&D Engineer *Toronto, ON*

- Led a cross-lab initiative across **9+ researchers and engineers** spanning neuroscience and hardware teams to evaluate transcranial focused ultrasound neuromodulation and MEG neural monitoring.
- Drove prototyping of a wearable neuromodulation helmet in SolidWorks, iterating through **11+ transducer/sensor configurations** and validating integration through phantom testing and simulation.
- Identified and mitigated electromagnetic interference using shielded coaxial cabling and signal filtering.
- Co-authored a **$225,000** internal research grant and coordinated technical priorities across **3 research groups**.

Vision & Image Processing Lab and Critical ML Lab Aug. 2025 – Present

Undergraduate Research Assistant *Waterloo, ON*

- Co-authored **DiCoH**, a self-supervised contrastive pretraining framework for homogeneous medical images (**under review**), improving on the strongest baseline by **2.0 mIoU** at 5% labels.
- Developing a **reinforcement learning–driven active learning** framework for label-efficient **semi-supervised segmentation**; leading reward-signal design using embedding-based selection criteria.
- Built a PyTorch active learning pipeline on **80K+ BraTS MRI slices**, adding spatially aware regional scoring and resolving a class-imbalance failure mode that biased sample selection.

Harvard Medical School – Tearney Lab May 2025 – Aug. 2025

Machine Learning Research Intern *Boston, MA*

- Built a **deep learning** pipeline from scratch for Parkinson's disease diagnosis via segmentation of esophageal **OCT** images, on a novel dataset targeting a signal never previously segmented.
- Trained by a neurologist to identify the signal; translated clinical feedback into **16+ design requirements** driving labeling, architecture, training, and evaluation.
- Trained and validated an **LT-U-Net** model, achieving over **85% classification accuracy** across all classes.
- Presented research at the **Wellman Center for Photomedicine** at Massachusetts General Hospital.

Projects

Myoelectric Prosthetic Arm | *Signal Processing, Circuit Design, EMG, C++, KiCad*

- Designed a **mixed-signal EMG acquisition and control system**, developing a 7-stage analog front-end for amplification, 20–450 Hz bandpass filtering, and digital triggering.
- Developed deterministic, non-blocking **C++ firmware** using a state machine to detect sustained muscle activation, control motor actuation, and implement timing-based trigger rejection.
- Designed a custom **KiCad PCB** integrating the analog front-end, microcontroller, and motor control.

Technical Skills

ML / Data: PyTorch, Computer Vision, Semantic Segmentation, Self-Supervised & Contrastive Learning, Active Learning, Reinforcement Learning
Programming: Python, C++, C#, Java, MATLAB, SQL
Signal Processing: MNE-Python, Digital Filter Design, Analog Signal Conditioning, Embedded DSP
Engineering: KiCad, SolidWorks, Git, Linux, PCB Design, Embedded Systems, Prototyping & Validation

Leadership

BioTEC Conference Oct. 2024 – Present

Logistics Director *Waterloo, ON*

- Managed **20+ executives** and led **logistics** for Canada's largest student-run biotech conference, with over **250 attendees**.